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POSTER++: A simpler and stronger facial expression recognition network

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ABSTRACT

The POSTER has achieved SOTA performance in facial expression recognition (FER) by effectively combining facial landmarks and image features through its two-stream pyramid cross-fusion design. However, the architecture of POSTER is undeniably complex. Specifically, the two-stream design and pyramidal architecture of POSTER involve a significant number of parameters. Moreover, the cross-attention mechanism used in POSTER's cross-fusion has a square-level computational complexity. These designs result in expensive computing cost for POSTER, which are not conducive to its industrial deployment in FER. To address the computational challenges associated with POSTER, this paper introduces POSTER++. POSTER++ simplifies the two-stream design employed in POSTER and introduces a lightweight multi-scale interaction (MSI) network as a replacement for POSTER's pyramidal architecture. MSI directly integrates facial information at different scales into the token embeddings through attentional interaction. This allows POSTER++ to allocate more attention to the token embedding at the scale that is most useful for the current facial image, thus improving discrimination. Furthermore, we propose a window-based cross-attention mechanism with linear complexity as an alternative to the vanilla cross-attention mechanism. Unlike Swin Transformer, POSTER++ uses single window query based on whole facial landmark features to interact with all windows of facial features, facilitating better capture of subtle changes in expression. Compared to POSTER, POSTER++ with the above improvements shows SOTA FER performance with less 39.1% parameters, 46.4% FLOPs, and 64.5% inference time. This proves the potential of POSTER++ as the backbone of FER industry deployment.

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1. Introduction

Facial expression recognition (FER) helps machines understand human emotions from facial expressions. This makes it a core task for human–computer interaction. In addition, with its powerful expression understanding ability, FER has great potential applications in psychology [1], product reviews, intelligent robotics, super-resolution, intelligent surveillance, virtual reality [2], and synthetic animation. In addition to deception detection efforts [3], facial expressions also play an important role in judging lying. Thus, research on FER is necessary.

Early FER works [4,5] used manual features for the analysis of facial expressions. However, FER algorithms based on manual features are often only applicable to specific FER tasks. When applied to real-world scenarios, it is difficult for these algorithms to achieve the same results as in the experimental setting. With the development of deep learning, convolutional neural networks (CNNs) [6] have been introduced to FER to improve the robustness of the network. Nevertheless, convolution-based FER algorithms are poor at considering the global information of the image due to the limitation of the convolutional local receptive field. Influenced by the vision transformer, Xue et al. [7] designed the first transformer-based FER network to model long-range dependencies for FER. Kim et al. [8] improved the vision transformer (ViT) to combine both global and local features so that ViT can be adapted to most FER tasks.

Among many excellent FER works, POSTER [9] stands out with SOTA performance by deals with the three key issues of FER simultaneously: inter-class similarity, intra-class discrepancy, and scale sensitivity. For these issues, POSTER proposes three core designs based on the characteristics of faces and facial landmarks: two-stream, cross-fusion, and pyramidal architecture. According to the difference and sparsity of facial landmarks, POSTER successfully solves inter-class similarity and intra-class discrepancy issues by cross-fusing facial features and facial landmark features in two branches of the two-stream design. The design of pyramid architecture introduces multi-scale features for POSTER to solve the scale sensitivity problem. Along with the solution to the three main FER issues, POSTER shows amazing expression analysis ability.

Although POSTER works well on FER, the large number of parameters and expensive computational cost brought by its network architecture prevent its industrial deployment in FER. In detail, the two-stream design and the pyramid structure of POSTER contain a huge number of parameters, and the vanilla cross-attention mechanism for fusing facial landmark features with facial features has a computational complexity at the square level.

As a solution to these problems, we improved the network design of POSTER and through extensive experimentation obtained a simpler, stronger POSTER++. For the two-stream design of POSTER, we evaluate each of its two branches theoretically and experimentally. We remove the branch that contains a large number of parameters and contribute little to solving the main FER issues. Motivated by [10], we propose a window-based cross-attention mechanism with linear complexity to address the challenge of square-level computational complexity in POSTER cross-fusion. Benefiting from the difference and sparsity of facial landmarks, instead of dividing facial landmark features into multiple windows as Swin Transformer does, POSTER++ employs whole facial landmark features as window queries to capture salient facial information associated with each facial feature window.

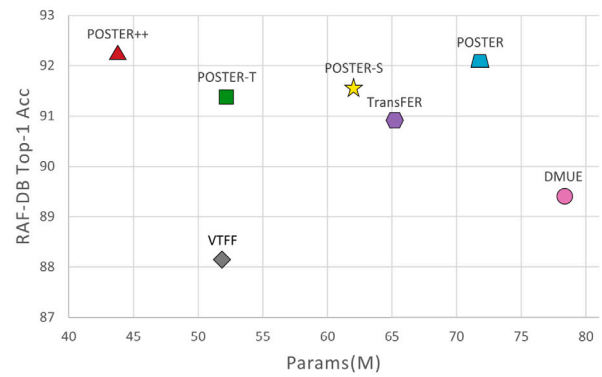

Fig. 1. POSTER++ results on RAF-DB. We compare POSTER++ with three variants of POSTER and other FER algorithms. The results indicate that POSTER++ weighs the number of parameters and accuracy better than other methods on RAF-DB.

Compared to the vanilla cross-attention mechanism, Window-based cross-attention mechanisms help to capture subtle changes in expressions. Meanwhile, with the window queries based on facial landmark features, POSTER++ can more precisely focus on salient areas of the face and thus more conducive to the resolution of FER inter-class similarity and intra-class discrepancy issues. For POSTER's pyramidal network, we propose a lightweight multi-scale interaction (MSI) network as its alternative. According to the attentional design, MSI network can extract multi-scale facial features from different scale token embeddings based on attention weights that are more beneficial to the current task. Based on these above designs, POSTER++ is simpler, faster, and stronger than POSTER in FER (see Fig. 1). Specifically, compared to POSTER, POSTER++ reduces parameters (Param) by 39.1%, the floating point operations (FLOPs) by 46.4%, and the inference time by 64.5%. Besides, It achieves SOTA performance on standard FER datasets with only 8.4G FLOPs and 43.7M Param.

Experimental results show POSTER++ reached 92.21% on RAF-DB [11], 67.49% on AffectNet [12] (7 cls), and 63.77% on AffectNet (8 cls). This is better than POSTER (RAF-DB with 92.05%, AffectNet (7 cls) with 67.31%, and AffectNet (8 cls) with 63.34%). POSTER++ offers a smaller Params (43.7M vs. 71.8M), FLOPs (8.4G vs. 15.7G), and inference time (9.1 ms vs. 25.7 ms). These results demonstrate the potential of POSTER++ as a solution for FER industrial deployment.

In general, we summarize the contributions of this paper as follows:

(1) We simplified POSTER and proposed a simpler, stronger, and faster FER network POSTER++.
(2) POSTER++ shows state-of-the-art performance on several standard FER datasets.
(3) POSTER++ reduces 39.1% parameters, 46.4% FLOPs, and 64.5% inference time of POSTER.

2. Related work

2.1. Facial expression recognition

The study of FER [13,14] has become very popular in recent years as more and more researchers focus on human–computer interaction.

Zhao et al. [15] used the manual feature LBP for the research of FER with good results. However, these manual representations lack robustness when applied to FER tasks in other scenes. Sang et al. [16] focused on reducing intra-class variation in facial expression features and introduced a dense convolutional network for the FER task. FER-net [17] introduces softmax classifiers for convolutional networks to efficiently distinguish expressions. Mohan et al. [18] introduced local gravitational force descriptor to facial expression recognition convolutional networks. While these methods improve the robustness of the FER model to some extent, the inherent shortcomings of convolution prevent them from considering the global structure.

With the increasing prominence of transformers in the field of computer vision, numerous FER methods incorporating transformers have emerged. Xue et al. [7] were the first to employ a vision transformer for FER research, achieving state-of-the-art performance. Despite the introduction of designs such as attention effectively improving expression recognition performance, current FER works still faces the challenges of intra-class variation, inter-class similarity, and scale sensitivity. Hence, some work [19] suggests combining facial action units to detect facial expressions. FLEPNet [20] reduces intra-class differences in facial expression images by fusing the combined features retrieved by the two networks. Mohan et al. [21] focused on five important local regions of the human face to emphasize expression-specific characteristics. POSTER [9] solves the intra-class discrepancy, inter-class similarity, and scale sensitivity issues of FER in the same time by integrating image features with facial landmark features through two-stream, cross-fusion, and pyramid design.

However, the huge computational cost of POSTER has prevented researchers from further research on FER. To solve this issue, we improved the architecture of POSTER and proposed POSTER++, which is simpler and more powerful for FER tasks.

2.2. Vision transformer

Recently vision transformers [22,23] have been widely used for computer vision tasks on large-scale datasets with their excellent ability to model long-distance dependencies.

Dosovitskiy et al. [24] pioneered the introduction of transformer from the field of natural language processing to computer vision. To make ViT outperform traditional ResNet on datasets without huge data size, Yuan et al. proposed T2T-ViT [25]. Chen et al. proposed CrossViT [26], which combines image patches of different sizes by dual branches to produce robust image features. Liu et al. [10] reduced the attention mechanism from quadratic computational complexity to linear by window attention and the design of a shift window scheme.

The architectural design of these vision transformer efforts inspires our improvements for POSTER [9]. This leads to a better trade-off between accuracy and computational complexity in FER with our POSTER++.

3. Method

In this section, we first review the POSTER process. We then describe the overall architecture of POSTER++ and discuss its three specific improvements to POSTER: two-stream simplification, window-based cross-attention mechanism, and multi-scale interaction network.

3.1. A brief review of POSTER

POSTER includes four modules: facial landmark detector, image backbone, cross-fusion transformer encoders, and pyramid network. Given the input image $X \in \mathbb{R}^{H\times W\times 3}$, POSTER obtains the image features X_{img} and landmark features X_{lm} by the image backbone and the facial landmark detector, respectively.

The image features $X_{img} \in \mathbb{R}^{N\times D}$ as well as the landmark features $X_{lm} \in \mathbb{R}^{N\times D}$ are mapped into three matrices: image query matrix Q_{img}, image key matrix K_{img}, image value matrix V_{img} and landmark query matrix Q_{lm}, landmark key matrix K_{lm}, and landmark value matrix V_{lm} in the cross-fusion transformer encoder. Specifically expressed as:

$$\begin{aligned} &Q_{img} = X_{img}W_{q1}, Q_{lm} = X_{lm}W_{q2}, \\ &K_{img} = X_{img}W_{k1}, K_{lm} = X_{lm}W_{k2}, \\ &V_{img} = X_{img}W_{v1}, V_{lm} = X_{lm}W_{v2}, \end{aligned} \tag{1}$$

where $W_{q1}, W_{q2}, W_{k1}, W_{k2}, W_{v1}$ and $W_{v2} \in \mathbb{R}^{D\times D}$ are the mapping matrices.

The cross-fusion transformer encoder primarily consists of two branches in the POSTER two-stream design: the image-to-landmark branch and the landmark-to-image branch. In both branches, the facial landmark features and facial features are cross-fused using the vanilla cross-attention mechanism. It is defined as follows:

$$\begin{aligned} &\text{Attention}_{(img)} = \text{softmax}(Q_{lm}K_{img}^{T}/\sqrt{d})V_{img}, \\ &\text{Attention}_{(lm)} = \text{softmax}(Q_{img}K_{lm}^{T}/\sqrt{d})V_{lm}, \end{aligned} \tag{2}$$

where softmax$(\cdot)$ is the softmax [27] activation function and $\frac{1}{\sqrt{d}}$ is an appropriately normalized scaling factor used to prevent the gradient from being too small.

In summary, the landmark-to-image branch is specifically represented as:

$$\begin{aligned} &X_{img}^{'} = \text{Attention}_{(img)} + X_{img}, \\ &X_{img_o} = \text{MLP}(\text{Norm}(X_{img}^{'})) + X_{img}^{'}, \end{aligned} \tag{3}$$

where MLP $(\cdot)$ is the Multi-Layer Perceptrons and Norm $(\cdot)$ denotes the normalization operation. The image-to-landmark branch is specifically denoted as:

$$\begin{aligned} &X_{lm}^{'} = \text{Attention}_{(lm)} + X_{lm}, \\ &X_{lm_o} = \text{MLP}(\text{Norm}(X_{lm}^{'})) + X_{lm}^{'}, \end{aligned} \tag{4}$$

Finally, POSTER extracts and integrates multi-scale features of images and landmarks by the pyramid network design. The specific details are shown in Fig. 2.

3.2. Architecture

As shown in Fig. 3, POSTER++ follows the pipeline of POSTER. POSTER++ still employs the facial landmark detector and image backbone in POSTER to extract facial landmark embeddings and facial embeddings. In contrast, POSTER++ extracts facial landmark embeddings and facial embeddings at different scales, respectively. This allows POSTER++ to avoid the high parameters of the POSTER pyramid network while having better multi-scale properties. For facial embedding and facial landmark embedding at different scales, POSTER++ only applies the landmark-to-image branch of the POSTER two-stream design for cross-fusion to reduce computational costs. The corresponding experimental and theoretical analyses also prove that this improvement is feasible. In the cross-fusion process, POSTER++ proposes a window-based cross-attention mechanism with linear computational complexity to replace the vanilla cross-attention mechanism with square-level computational complexity. The overall facial landmark feature is used by POSTER++ as a single window query to interact with each window of the facial feature to capture more detailed and significant facial features to better address inter-class similarity and intra-class variability issues in FER. After cross-fusion, POSTER++ merges the obtained different scale token embeddings and inputs them uniformly into our proposed lightweight MSI network for attention computation. For different facial image inputs, MSI can dynamically adjust the attention weights. POSTER++ weights the token embeddings at different scales according to the attention weights provided by MSI to better meet the expression analysis in different scale situations.

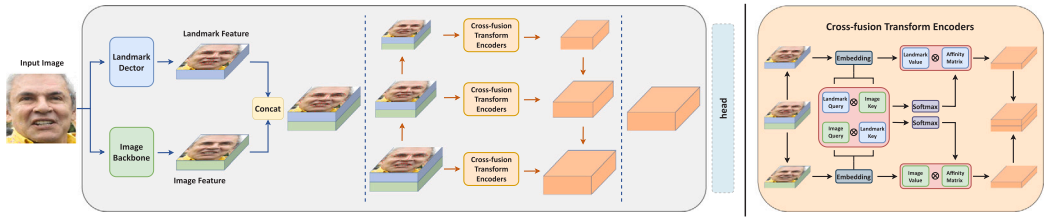

Fig. 2. Pipeline of POSTER. POSTER mainly contains facial landmark detector, image backbone, cross-fusion transformer encoders and pyramid network.

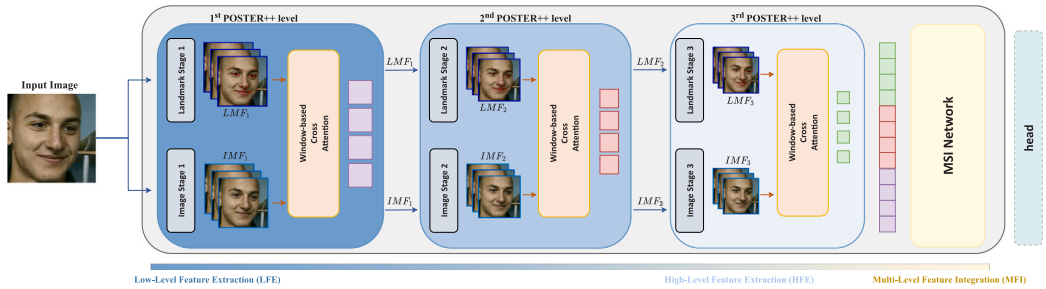

Fig. 3. The overview of POSTER++ architecture. LMF_i and IMF_i denotes facial landmark features and image features at the ith level of POSTER++ respectively. Level indicates the different scales of the feature.

Table 1
Ablation study of two branches in cross-fusion of POSTER. The baseline in the table keeps the baseline setting in POSTER.

Methods	RAF-DB
Baseline	91.00
POSTER	92.05
POSTER w/o image-to-landmark branch	91.82
POSTER w/o landmark-to-image branch	91.62

3.3. Two-stream simplification

To reduce the parameters in the two-stream design of the POSTER, we carefully examine its internal architecture. Although two-stream is central to the design of POSTER, POSTER does not explore which branch of two-stream actually plays a major role. Thus, in this section, we first perform an ablation study of the two-stream to learn the contribution of the two branches to the FER.

Table 1 shows the ablation results on the RAF-DB dataset, the accuracy of POSTER slips by 0.23 after missing the image-to-landmark branch. If the landmark-to-image branch is missing, the accuracy of POSTER on RAF-DB is reduced by 0.43. These findings suggest that while the image-to-landmark branch contributes to the overall FER performance of POSTER, it is the landmark-to-image branch that plays a crucial role in determining the effectiveness of POSTER. Next, we analyze the above results at the theoretical level.

Discussion. The two-stream design in POSTER is used to solve the intra-class discrepancy and inter-class similarity issues in FER. It includes landmark-to-image and image-to-landmark branches. We revisit the influence of the two branches on POSTER. In the landmark-to-image branch, the landmark features interact with the image features as queries Q_{lm} in the cross-attention mechanism. Image features are guided by landmark features to more easily represent facial expression salient regions of facial expressions when dealing with intra-class discrepancy issues. Additionally, benefiting from the sparsity of landmark features, image features guided by landmark features reduce the focus on regions where faces are prevalent. This helps to reduce the impact of inter-class similarity in FER. The results of the visualization of landmark-to-image branch attention in Fig. 4 also validate the above statements. Therefore, the landmark-to-image branch in the two-stream is essential and needs to be retained. For the image-to-landmark branch, the image features interact with the landmark features as queries Q_{img} to compensate for the lack of landmark features. Although this also benefits the FER task to some extent, it does not contribute to solving the issues of inter-class similarity and intra-class discrepancy and comes with a large computational cost. Since POSTER++ is designed to solve the FER issues that POSTER has to process while reducing its computational consumption. Thus, by making a trade-off between computational cost and accuracy of image-to-landmark branch, we eventually remove the image-to-landmark branch in the two-stream design.

3.4. Window-based cross-attention mechanism

For the square-level computational overhead of the POSTER cross-fusion process, we design a window-based cross-attention mechanism instead of the vanilla cross-attention mechanism in POSTER for linear computation in POSTER++. Meanwhile, in order to make the tokens of each window of facial features be guided by all facial landmark features, we do not adopt the window attention mechanism in Swin Transformer [10] but propose a new window-based cross-attention strategy. Fig. 5 illustrates the detailed differences among the three cross-attention mechanisms. For image features $X_{img} \in \mathbb{R}^{N\times D}$, we first divide them into several nonoverlapping windows $z_{img} \in \mathbb{R}^{M\times D}$, where z_{img} contains M tokens. For the landmark feature $X_{lm} \in \mathbb{R}^{C\times H\times W}$, we first downsample it to the window size $z_{lm} \in \mathbb{R}^{c\times h\times w}$, where $c = D, M = h\times w$. Then we reshape it into a single window according to the shape of z_{img}. At this point, the cross-attention with I heads in a local window can be formulated as:

$$\begin{aligned} &q = z_{lm}w_q, k = z_{img}w_k, v = z_{img}w_v, \\ &o^{(i)} = \mathrm{softmax}(q^{(i)}k^{(i)T}/\sqrt{d} + b)v^{(i)}, i = 1, \dots, I, \\ &o = [o^{(1)}, \dots, o^{(I)}]w_o, \end{aligned} \tag{5}$$

where w_q, w_k, w_v, w_o are the mapping matrices. $\theta(\cdot)$ is the softmax function. $[\cdot]$ denotes the merge operation and $b \in \mathbb{R}^{I\times I}$ is the relative position bias.

We perform the above cross-attention calculation for all windows. We refer to this cross-attention mechanism as window-based multihead cross-attention (W-MCSA). The cross-fusion transformer encoder in POSTER++ can be expressed as follows:

$$\begin{aligned} &X'_{img} = \text{W-MCSA}(img) + X_{img}, \\ &X_{img_o} = \mathrm{MLP}(\mathrm{Norm}(X'_{img})) + X'_{img}, \end{aligned} \tag{6}$$

Fig. 4. Input images (row 1), facial landmark images (row 2), landmark-to-image branching attention visualization results (row 3). We visualize the attention map belonging to the last layer of the landmarks to image branching for high-level features in POSTER. We can observe that with the help of landmark features, the attention map focuses more on the outstanding areas of face and less on the areas common to face.

Fig. 5. Window-based cross-attention mechanism (ours), vanilla cross-attention mechanism (ViT), and Window cross-attention mechanism (Swin Transformer).

Based on the cross-attention between the whole facial landmark feature window query and each window of the facial features, POSTER++ can more effectively exploit the sparsity and saliency of facial landmarks to reduce the attention paid to some universal facial representations and thus more accurately capture salient features between each region of the face. This makes POSTER++ more favorable for solving the issues of inter-class similarity and intra-class discrepancy of FER. Finally, our proposed multi-scale interaction network will establish communication for tokens between different windows.

Computational complexity analysis. Since the query in the two types of cross-attention computation keeps the same shape as the key, value, we can use the multihead self-attention and the window-based multihead self-attention complexity in Swin Transformer [10] to represent their computational complexity. This can be indicated as follows:

$$\begin{aligned} \Omega(\text{MCSA}) &= 4ND^2 + 2N^2D, \\ \Omega(\text{W-MCSA}) &= 4ND^2 + 2M^2ND, \end{aligned} \tag{7}$$

According to Eq. (7), we find that our window-based cross-attention mechanism successfully reduces the computational complexity of cross-fusion in POSTER from the square level to the linear level. This further improves the computational efficiency of POSTER++.

3.5. Multi-scale interaction network

For the vast parameters in the POSTER pyramid architecture, POSTER++ proposes the MSI network as a solution. In POSTER++, we no longer use the pyramid architecture to address the scale sensitivity issue in FER. POSTER++ extracts facial and landmark embeddings at different scales directly from the facial landmark detector and the image backbone. Directly extracted multi-scale embeddings offer a wider choice of scales for scale sensitivity issues in FER than pyramidal architectures. For the multi-scale token embeddings o^1, o^2, and o^3 obtained after cross-fusion, POSTER++ merges them directly in the token dimension and performs the attention calculation with a lightweight network MSI consisting of only two layers of vanilla transformer blocks:

$$\begin{aligned} p &= [o^1, o^2, o^3], \\ p &= \text{MSI}(p), \end{aligned} \tag{8}$$

where $\text{MSI}(\cdot)$ represents multi-scale interaction network. Based on the attention weights provided by the MSI, POSTER++ can select different scale token embeddings suitable for the current task in order to better achieve the expression analysis. For the above design, we discuss as follows.

Discussion. POSTER adopts the pyramid structure to solve the scale sensitivity problem in FER. However, we consider that the pyramid structure design is only an upsampling and downsampling operation based on the same scale feature map. While it provides some degree of multi-scale information, we believe that directly extracting multi-scale features from the network is a more effective approach, which offers a wider choice of scales for scale-sensitive problem. For the integration and interaction of multi-scale features, the MSI network consisting only of vanilla transformer blocks are sufficient for this task. Through the attention mechanism, the MSI network can establish long-range dependencies for token embeddings at all scales and thus perform efficient aggregation of token embeddings at different scales. Based on the above design, the MSI network can provide more weights for the token embeddings appropriate to the current facial image scale. This

Table 2
Effects of different frameworks and different scales of facial features on POSTER++.

Methods	RAF-DB	AffectNet
w/ MLP	91.88	66.98
w/ POSTER pyramid architecture	83.73	61.59
w/ 1st facial feature	79.69	52.25
w/ 2st facial feature	79.79	52.97
w/ 3st facial feature	81.38	59.79
w/ MSI network	92.21	67.49

Table 3
Detailed size of the experimental dataset.

Dataset	Train size	Test size	Classes
RAF-DB [11]	12 271	3068	7
AffectNet (7 cls) [12]	280 401	3500	7
AffectNet (8 cls) [12]	283 501	4000	8
CAER-S [28]	44 996	20 987	7

Table 4
Supervised training POSTER++ from scratch on RAF-DB.

Config	Value
optimizer	Adam
base learning rate	3.50E−05
weight decay	1.00E−04
batch size	144
training epochs	200
learning rate schedule	ExponentialLR (gamma=0.98)
augmentation	RandomHorizontalFlip(), RandomErasing(scale=(0.02, 0.1)).
drop path	linspace(0, 0.5, 5)
num classes	7

adaptive multi-scale token selection approach provides more schemes and fewer parameters for the solution of FER scale sensitivity issue than pyramidal networks.

Additionally, we report the results of different frameworks or different scale facial features on POSTER++. For the architecture, we replace the MSI network with 3 layers of MLP or pyramid architecture in POSTER. Table 4 represents MSI network is more suitable for feature aggregation of different scale facial features. Besides, we also test the result of sole facial features of different scales in Table 2. These experiments jointly prove that our MSI network can better focus on the token embeddings whose scale is more appropriate to the specific facial image.

4. Experiments

We verify the effectiveness of POSTER++ on several standard FER datasets such as RAF-DB [11], AffectNet [12], and CAER-S [28]. In the following, we first compare POSTER++ with SOTA FER methods. A series of method analyses and ablation studies are then applied to POSTER++.

4.1. Experimental setup

Datasets. We evaluate the FER performance of POSTER++ on the widely used RAF-DB, AffectNet, and CAER-S datasets. The RAF-DB is a large-scale database of facial expressions. For the selection of expressions, RAF-DB selected six basic emotions as well as neutral emotions from a range of expressions for a total of seven expressions for expression annotation. AffectNet is currently the largest publicly available dataset in the FER field. It mainly contains 8 categories of primary emotions (neutral, happy, angry, sad, fear, surprise, disgust, contempt). The CAER-S dataset was obtained from the CAER dataset containing 65,983 images. It is mainly divided into 7 types of expressions: neutral, happy, sad, surprised, fear, disgust, and anger. The specific dataset configuration is shown in Table 3.

Table 5
Detailed information on facial landmark representations at different levels.

IMF_i	MobileFaceNet level	shape
i = 1	3	(28,28,64)
i = 2	4	(14,14,128)
i = 3	6	(7,7,512)

Table 6
Supervised training POSTER++ from scratch on AffectNet (7 cls).

Config	Value
optimizer	Adam
base learning rate	1.00E−06
weight decay	1.00E−04
batch size	144
training epochs	200
learning rate schedule	ExponentialLR (gamma=0.98)
augmentation	RandomHorizontalFlip(), RandomErasing(p=1, scale=(0.05, 0.05)).
drop path	linspace(0, 0.5, 5)
num classes	7

Settings. Similar to POSTER [9], we also use the IR-50 [29] network pretrained on the Ms-Celeb-1M dataset as the image backbone. MobileFaceNet [30] with frozen weights is used as our facial landmark detector. POSTER++ extracts the latent representations of facial expression images in MobileFaceNet at layers 3, 4, and 6 as facial landmark features at the first, second, and third levels of POSTER++, respectively, and the shapes of specific facial landmark features are shown in Table 5. Meanwhile, we employ the Adam optimizer for 200 epochs training. A training scheme with a batch size of 144, a learning rate of 3.5e−4, and a weight decay of 1e-4 was used. We use random horizontal flipping and random erasing as our data augmentation methods. For the loss function, we choose the standard cross-entropy loss. We eventually realize POSTER++ on a single NVIDIA RTX 3090 via PyTorch.

4.2. Implementation details

For POSTER++, we conduct FER experiments on the RAF-DB, AffectNet, and CAER-S datasets, respectively. For different datasets, we adopt different detail settings. Specifically, for different datasets, we exploit different learning rates for training according to the settings of POSTER. Moreover, for AffectNet (8 cls), POSTER++ uses a classification head with a category number of 8 for prediction. The rest of the settings are consistent with the experimental sections in the main text.

RAF-DB settings. We use the Adam optimizer with a learning rate of 3.5e−5 for 200 epochs training. The batch size is maintained at 144 and the weight decay remains at 1e-4. The learning rate schedule uses an exponential decay with a gamma of 0.98. Data augmentation includes random horizontal flipping and random erasure. The specific settings are shown in Table 4.

AffectNet (7 cls) settings. On the AffcetNet (7 cls) dataset, we adjust the learning rate to 1e-6. The training epochs remain at 200. The batch size is maintained at 144 and the weight decay remains at 1e-4. The learning rate schedule uses an exponential decay with a gamma of 0.98. Data augmentation includes random horizontal flipping and random erasure. The detailed settings are shown in Table 6.

AffectNet (8 cls) settings. We use the Adam optimizer with a learning rate of 1e-6 for 200 epochs training. The batch size is maintained at 144 and the weight decay remains at 1e-4. The learning rate schedule uses an exponential decay with a gamma of 0.98. Data augmentation

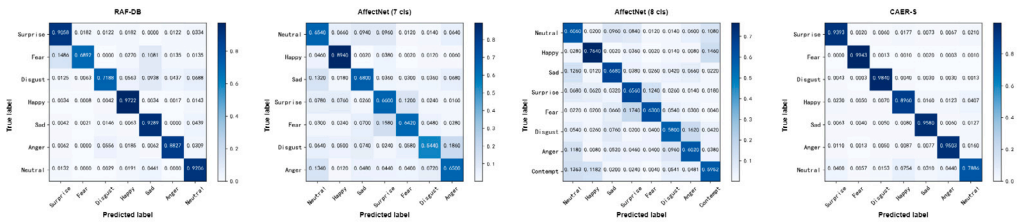

Fig. 6. The confusion matrix of POSTER++ on each dataset.

Table 7
Supervised training POSTER++ from scratch on AffectNet (8 cls).

Config	Value
optimizer	Adam
base learning rate	1.00E−06
weight decay	1.00E−04
batch size	144
training epochs	200
learning rate schedule	ExponentialLR (gamma=0.98)
augmentation	RandomHorizontalFlip(), RandomErasing(p=1, scale=(0.05, 0.05)).
drop path	linspace(0, 0.5, 5)
num classes	8

Table 8
Supervised training POSTER++ from scratch on CAER-S.

Config	Value
optimizer	Adam
base learning rate	4.00E−05
weight decay	1.00E−04
batch size	144
training epochs	200
learning rate schedule	ExponentialLR (gamma=0.98)
augmentation	RandomHorizontalFlip(), RandomErasing(p=1, scale=(0.05, 0.05)).
drop path	linspace(0, 0.5, 5)
num classes	7

includes random horizontal flipping and random erasure. In addition, we set the number of categories to 8. Table 7 shows the specific experimental settings.

CAER-S settings. On the CAER-S dataset, we employ the Adam optimizer with a learning rate of 4e-5 for 200 epochs of training. The batch size is maintained at 144 and the weight decay remains at 1e-4. The learning rate schedule uses an exponential decay with a gamma of 0.98. Data augmentation includes random horizontal flipping and random erasure. The specific settings are shown in Table 8.

4.3. Comparison with SOTA FER methods

Results on RAF-DB. We compare POSTER++ with the SOTA FER algorithms in recent years on the RAF-DB datasets in Table 10. The experimental results show that POSTER++ exhibits SOTA performance on RAF-DB. Compared with POSTER (92.05), POSTER++ improved by 0.16. +1.86 for POSTER++ over EAC (90.35), and +1.3 for POSTER++ over TransFER (90.91). Table 9 shows the comparison of POSTER++ with POSTER for RAF-DB individual classes and average accuracy. Although POSTER++ outperformed POSTER in several categories, the average accuracy was slightly inferior to that of POSTER (see Fig. 6 for the confusion matrix on RAF-DB).

Results on AffectNet. In Table 10, we also conduct FER experiments on AffectNet (7 cls) as well as AffectNet (8 cls). We observe that POSTER++ exhibit SOTA FER results in both AffectNet (7 cls) and AffectNet (8 cls). Compared with POSTER (67.31, 63.34), POSTER++ increases by 0.18 and 0.43 on AffectNet (7 cls) and AffectNet (8 cls), respectively. On AffectNet (8 cls), POSTER++ is higher than DAN (62.09) by 1.68. On AffectNet (7 cls), POSTER++ is greater than TransFER (66.23) with 1.26. This demonstrates that POSTER++ can maintain excellent FER performance even on larger datasets. Table 9 shows that POSTER++ exceeds POSTER for the majority of individual class accuracies in both AffectNet (7 cls) and AffectNet (8 cls). As a result, POSTER++ achieves better average accuracy than POSTER on AffectNet (see Fig. 6 for the confusion matrix on AffectNet).

Results on CAER-S. We compare POSTER++ with recent SOTA FER methods on the CAER-S dataset. POSTER++ in Table 11 performs extremely well on the CAER-S dataset. Specifically, POSTER++ scored 92.98 on the CAER-S. +0.27 for POSTER++ over POSTER (92.73). +3.12 for POSTER++ over GLAMOR-Net (89.88) and +4.58 for POSTER++ over MA-Net (88.42). +7.13 for POSTER++ over Efficient-Face (85.87). The excellent results on CAER-S prove that the success of POSTER++ is no accident (see Fig. 6 for the confusion matrix on CAER-S).

Statistical hypothesis test. Statistical hypothesis tests were performed to more intuitively reflect the difference in performance between POSTER++ and other methods. Specifically, we first formulate the original hypothesis (H_0) that POSTER++ does not perform significantly better than the baseline and the alternative hypothesis (H_1) that POSTER++ performs significantly better than the baseline. Then we introduce the accuracies of POSTER++ and the comparison baseline on the test set and compute their differences, finally, we perform a paired T-test to obtain the corresponding t-statistics and p-values obtained for POSTER++ and the comparison baseline. The hypothesis is rejected when the p-value is <0.05, otherwise the hypothesis is valid. a larger t-statistic indicates a larger difference between the two sample means.

Table 12 demonstrates the paired T-test results. Based on the statistical hypothesis test results, it can be found that there is a significant difference in the performance of POSTER++ and the baselines. This implies that POSTER++ significantly outperforms the baselines on these datasets.

4.4. Comparison of computational efficiency

From Table 13, we can see that POSTER++ achieves better FER results with smaller Params, FLOPs, and inference time than baseline. POSTER++ further improves the performance of the FER system with higher computational efficiency compared to the work of POSTER, TransFER, DMUE, and SwinFace. Although our method has higher FLOPs compared to MA-Net, POSTER++ exhibits higher real-time and performance. DACL exhibits lower computational cost and higher real-time performance on FER tasks, but its classification results still have some gaps from POSTER++. Compared to POSTER-T, POSTER++ reduces 8.5M Params, 5.2G FLOPs, and 7.4 ms, while increasing 0.85% on RAF-DB and 0.62% on AffectNet. Compared to POSTER-S, POSTER++ reduces 18.3M Params, 6.3G FLOPs, and 12.1 ms, while increasing 0.67% on RAF-DB and 0.36% on AffectNet. Compared to POSTER, POSTER++ reduces 28.1M Params, 7.3G FLOPs, and 16.6 ms, while increasing 0.16% on RAF-DB and 0.18% on AffectNet. Compared to these baselines, POSTER++ achieves a better trade-off for computational cost, real-time, and FER performance. Therefore, POSTER++ is a better choice for industrial deployments of FER.

Table 9
Class-wise accuracy of POSTER and POSTER++ on RAF-DB, AffectNet (7 cls), and AffectNet (8 cls) datasets. Green, blue, and red mark the highest value of single category in RAF-DB, AffectNet (7 cls), and AffectNet (8 cls) respectively.

Dataset	Method	Neutral	Happy	Sad	Surprise	Fear	Disgust	Anger	Contempt	mean Acc
RAF-DB	POSTER [9]	92.35	96.96	91.21	90.27	67.57	75.00	88.89	–	**86.04**
RAF-DB	SqueezExpNet [31]	75.62	75.28	73.82	74.60	47.53	68.61	69.48	–	69.28
RAF-DB	DCNN+RLPS [32]	70.91	91.55	51.81	68.68	42.20	33.22	57.00	–	59.34
RAF-DB	POSTER++	92.06	97.22	92.89	90.58	68.92	71.88	88.27	–	85.97
AffectNet (7 cls)	POSTER [9]	67.20	89.00	67.00	64.00	64.80	56.00	62.60	–	67.23
AffectNet (7 cls)	POSTER++	65.40	89.40	68.00	66.00	64.20	54.40	65.00	–	**67.45**
AffectNet (8 cls)	POSTER [9]	59.40	80.20	66.60	63.60	63.60	59.80	58.80	54.71	63.34
AffectNet (8 cls)	POSTER++	60.60	76.40	66.80	65.60	63.00	58.00	60.20	59.52	**63.76**

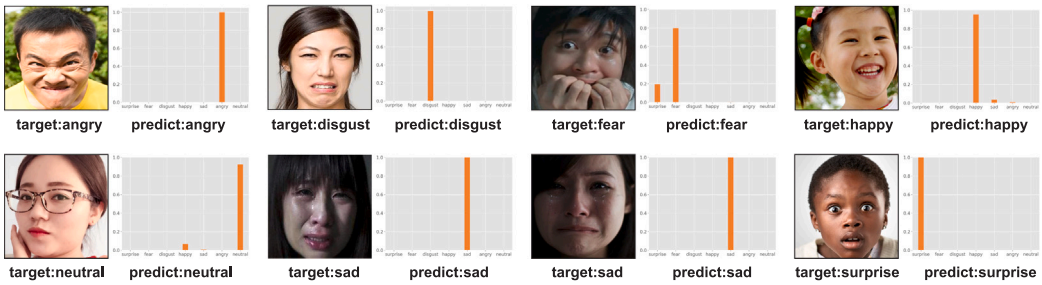

Fig. 7. Analysis results of POSTER++ for internet facial expression images.

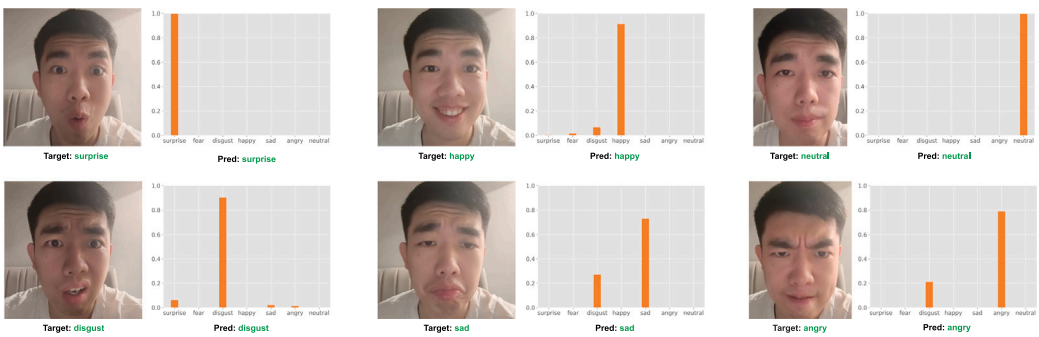

Fig. 8. Analysis results of POSTER++ for facial expression images of the same identity taken by mobile phone.

4.5. Practical application

To further validate the performance of POSTER++ in real-world scenarios, we collected several real facial expression images from the internet for testing. Meanwhile, we took several real facial images with a mobile phone to evaluate the performance of POSTER++ in recognizing different expressions of the same face in the real world.

Fig. 7 shows that POSTER++ can accurately analyze a wide range of facial expression images in real-world scenarios, while it can make confident judgments for most of the facial expression images in real-world scenarios.

Meanwhile, Fig. 8 indicates that POSTER++ can also achieve accurate recognition for different expressions of the same face in the real world, which also shows that POSTER++ can effectively deal with the challenges of inter-class similarity and intra-class variability of the FER task in real-world scenarios.

4.6. Method analysis

In this subsection, we present a method analysis for the MSI network we used in POSTER++ on RAF-DB.

MSI network depth. Here, we investigate the impact of different depths for the MSI network on the FER performance of POSTER++. In Fig. 9, we show the influence of the MSI network with depth {2,4,6,8} on POSTER++. We observe that for multi-scale integration and interaction, we do not need to increase the depth of the MSI network. The MSI network with a depth of 2 is sufficient to handle the FER task. A deeper MSI network hurts the performance of POSTER++ instead. A deeper MSI Network might cause severe overfitting issue in POSTER++.

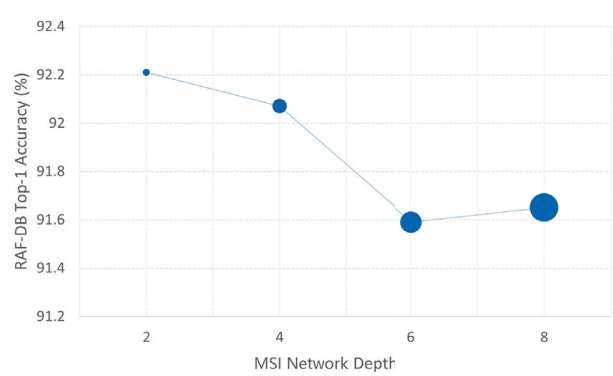

Fig. 9. Influence of different depth MSI networks on POSTER++ for RAF-DB.

Pretrained MSI network. We study the influence of the pre-trained MSI network on POSTER++. For the MSI network, we utilized the

Table 10
Comparison results with SOTA FER algorithms on RAF-DB and AffectNet.

Methods	Year	RAF-DB	AffectNet (7 cls)	AffectNet (8 cls)
PSR [13]	2020	88.98	63.77	60.68
DACL [33]	2020	87.78	65.20	–
DMUE [34]	2021	89.42	63.11	–
VTFF [14]	2021	88.14	61.85	–
ARM [35]	2021	90.42	65.20	61.33
TransFER [7]	2021	90.91	66.23	–
DAN [36]	2021	89.70	65.69	62.09
EfficientFace [6]	2021	88.36	63.70	60.23
MA-Net [37]	2021	88.42	64.53	60.29
Zhang et al. [38]	2022	87.10	62.10	–
Meta-Face2Exp [39]	2022	88.54	64.23	–
EAC [40]	2022	90.35	65.32	–
AMP-Net [41]	2022	89.25	64.54	61.74
FER-CHC [42]	2023	90.06	66.31	–
FST-MWOS [43]	2023	90.38	63.33	–
SwinFace [44]	2023	90.97	–	–
MACN [45]	2023	90.16	63.12	–
Face2Nodes [46]	2023	91.02	66.69	–
MRAN [47]	2023	90.03	66.31	62.48
SqueezExpNet [31]	2023	80.65	–	–
SSA-ICL-ResNet18 [48]	2023	89.44	65.78	–
POSTER [9]	2023	92.05	67.31	63.34
HAM [49]	2024	91.92	66.97	63.82
Wang et al. [50]	2024	–	62.70	–
POSTER++	–	92.21	67.49	63.77

Table 11
Comparison results with SOTA FER method on CAER-S.

Methods	Year	CAER-S
DSN [51]	ICML 2018	75.19
CAER-Net-S [28]	ICCV 2019	73.51
GRERN [52]	IEEE Access 2020	81.31
EfficientFace [6]	AAAI 2021	85.87
MA-Net [37]	TIP 2021	88.42
GLAMOR-Net [53]	NCA 2021	89.88
POSTER [9]	ICCVW 2023	92.73
SMResNet [54]	2023	88.52
HAM [49]	NN 2024	92.86
POSTER++	–	93.00

Table 12
Paired T-test results for POSTER++ vs. baseline.

Methods	t-statistic	p-value
PSR [13]	17.523	0.003
ARM [35]	11.060	0.008
TransFER [7]	64.0000	0.010
DAN [36]	7.709	0.016
EfficientFace [6]	5.363	0.013
MA-Net [37]	10.924	0.002
AMP-Net [41]	8.583	0.013
MRAN [47]	4.896	0.039
EAC [40]	13.000	0.049
POSTER [9]	4.227	0.024

pre-trained ViT weights from ImagenNet-21K. The performance of POSTER++ on FER (Facial Expression Recognition) demonstrated a decline subsequent to the incorporation of the pre-trained weights, as indicated in Table 14. We argue that this is mainly because the pre-trained MSI network acts on the feature extraction of the image-level inputs. However, in POSTER++, MSI network performs the multi-scale feature integration and interaction task of feature-level inputs. The difference in input and task resulted in the MSI network with pre-trained weights not working on POSTER++.

4.7. Ablation study

We validate the effectiveness of our POSTER improvement component on the RAF-DB and AffectNet datasets.

Multi-scale feature extraction. We first verify the effectiveness of extracting multi-scale features directly. In this ablation experiment, we only use the last layer of feature maps from the image backbone as well as the facial landmark detector for cross-fusion. From Table 15 we observe that POSTER++ degrades significantly on the RAF-DB and AffectNet datasets when multi-scale feature extraction is not performed. This shows that our method of directly extracting multi-scale features can also solve the scale sensitivity issue of FER. Additionally, this indicates the importance of multi-scale features for FER.

MSI network. For the MSI network used for multi-scale feature integration and interaction, we ablate it. We directly sum several different scale features for FER. The experimental results presented in Table 15 reveal a decrease in the performance of POSTER++ by 0.35 on the RAF-DB dataset and 0.57 on the AffectNet dataset when the MSI network is not involved in multi-scale feature integration and interaction. This suggests that the MSI network facilitates multi-scale feature integration.

W-MCSA. We validate the effectiveness of W-MCSA for cross-fusion by ablation experiments. In this experiment, we use the vanilla cross-attention mechanism to replace our W-MCSA. We observed that POSTER++ degraded on both the RAF-DB and AffectNet datasets. This shows that the W-MCSA we use both improves the FER accuracy and reduces the computational complexity of POSTER. In addition, we compare the efficiency of the attention scheme in POSTER++ and the window attention in Swin Transformer in facial expression recognition. In this ablation experiment, we replace the W-MSA in POSTER++ with the window attention mechanism in Swin Transformer. We divide both facial landmark features and facial features into several non-overlapping windows and compute the window cross-attention for tokens in the corresponding windows for different features. It can be noted that a degradation in FER performance occurs due to the lack of a global comprehension of facial landmark features in facial features (see Table 15).

Cross-fusion. This experiment verifies the role of the landmark-to-image branch for POSTER++. In the ablation experiments on cross-fusion, we merge the extracted image multi-scale features and landmark multi-scale features directly and integrate them by MSI. Table 15 shows that the effectiveness of POSTER++ on RAF-DB and AffectNet drops sharply when cross-fusion is not applied. This shows that cross-fusion is the key for POSTER++ to achieve SOTA FER. Furthermore, it indicates that addressing inter-class similarity and intra-class discrepancy is particularly important in FER tasks.

5. Visualization

T-SNE visualization. We visualized the high-dimensional features of POSTER and POSTER++ using t-SNE. As can be seen in Fig. 10, both POSTER++ and POSTER present good t-SNE visualization results on RAF-DB and CAER-S datasets. There is almost no significant difference between the t-SNE visualization results of POSTER and POSTER++ on CAER-S. POSTER++ has a closer intra-class distance than POSTER on RAF-DB. Although POSTER and POSTER++ have poor t-SNE visualization results on AffectNet (7 cls) and AffectNet (8 cls). However, the inter-class distance between clusters in POSTER++ is further than POSTER. The above results indicate that POSTER++ is better than POSTER in alleviating the issues of inter-class similarity and intra-class discrepancy in FER.

Attention visualization. We visualize the attention map of the highest-level features of the POSTER++ cross-fusion stage. From Fig. 11, we observe that POSTER++ successfully captures important facial expression features with the help of facial landmark features.

Table 13
Comparison of Param, FLOPs, FPS, and Inference Time with other methods.

Methods	#Params	#FLOPs	#Time	#FPS	RAF-DB (%)	AffectNet (%)
MA-Net [37]	63.5M	3.7G	17.6 ms	56.7 fps	88.42	64.53
DACL [33]	103.0M	1.9G	4.2 ms	238.1 fps	87.78	65.20
VTFF [14]	51.8M	–	–	–	88.14	61.85
FST-MWOS [43]	147.9M	33.82G	–	–	90.38	63.33
PACVT [55]	45.9M	–	–	–	87.18	59.76
TransFER [7]	65.2M	15.3G	12.2 ms	81.9 fps	90.91	66.23
DMUE [34]	78.4M	13.4G	14.5 ms	68.9 fps	89.42	63.11
HAM [49]	69.63M	–	–	–	91.92	66.97
SwinFace [44]	67.3M	23.8G	12.1 ms	82.6 fps	90.97	–
POSTER-T [9]	52.2M	13.6G	16.5 ms	60.6 fps	91.36	66.87
POSTER-S [9]	62.0M	14.7G	21.2 ms	47.2 fps	91.54	67.13
POSTER [9]	71.8M	15.7G	25.7 ms	38.9 fps	92.05	67.31
POSTER++	43.7M	8.4G	9.1 ms	109.9 fps	92.21	67.49

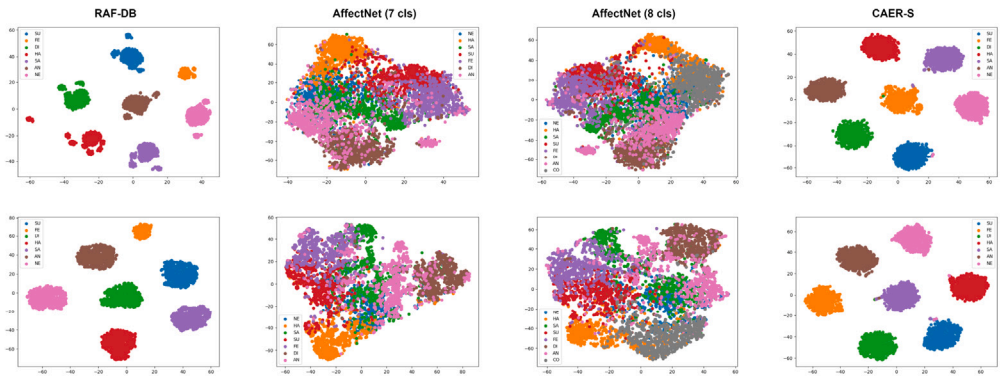

Fig. 10. Comparison of POSTER++ and POSTER high-dimensional space t-SNE visualization results. POSTER t-SNE visualization results (first row), POSTER++ t-SNE visualization results (second row).

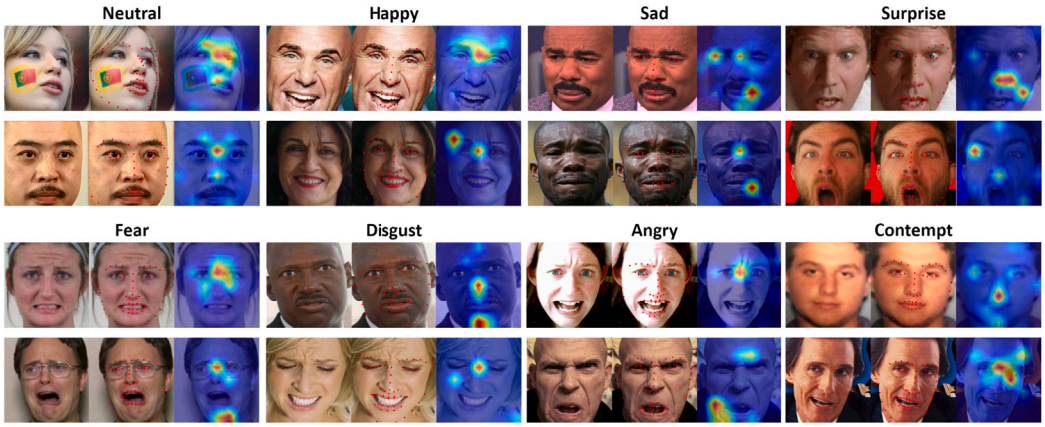

Fig. 11. POSTER++ cross-fusion stage attention visualization results. For each triplet, we show the input image (left), the landmark image (middle), and the attention map (right).

Table 14
Impact of pre-trained MSI network for POSTER++ on FER.

w/ pre-trained MSI network	RAF-DB	AffectNet
✓	91.49	60.20
✗	92.21	67.49

Table 15
Results of ablation experiments of key components of POSTER++.

Methods	RAF-DB	AffectNet
POSTER++	92.21	67.49
w/o multi-scale feature extraction	91.47	66.51
w/o MSI network	91.86	66.92
w/o W-MCSA (w/ vanilla cross-attention)	91.56	67.24
w/o W-MCSA (w/ window cross-attention)	91.27	66.40
w/o cross-fusion	91.39	66.35

6. Conclusion

In this paper, we simplify the two-stream design of POSTER and propose a window-based cross-attention mechanism and MSI to replace the vanilla cross-attention mechanism and pyramid network, thus removing the limitations of POSTER's huge number of parameters and computational complexity. Based on these improvements, we propose a simpler, stronger, and faster FER network, POSTER++. Extensive FER experimental results show that POSTER++ retains the advantages of POSTER on the FER task by combining facial representations and facial landmark representations successfully solving the

challenges of interclass similarity, intraclass variability, and scale sensitivity that are prevalent in current facial expression recognition tasks. Thus POSTER++ achieves SOTA FER performance while greatly reducing the Params, FLOPs, and inference time of POSTER. This suggests that POSTER++ achieves a better trade-off between accuracy and computational complexity. Thus, it is a better choice for FER industrial deployments. The success of POSTER++ means that the current stage of FER work can further consider more efficient network architecture design and introduce some novel facial feature designs related to expressions. However, in practical applications, some facial images have object occlusion, which may result in facial landmarks not being effectively detected thus further impairing the facial expression recognition performance of POSTER++. While the multi-scale design of the MSI network mitigates the occlusion issue to some extent, there is no specific design for object occlusion in POSTER++. In future work, we plan to employ POSTER++ as the backbone for designing masked autoencoders specific to facial expression recognition to learn occlusion-invariant representations for facial images to solve occlusion issue and further improve the training efficiency.

CRediT authorship contribution statement

Jiawei Mao: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Rui Xu:** Visualization, Software, Resources, Investigation, Data curation. **Xuesong Yin:** Writing – review & editing, Supervision. **Yuanqi Chang:** Visualization, Validation, Software, Data curation. **Binling Nie:** Writing – review & editing. **Aibin Huang:** Supervision. **Yigang Wang:** Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xuesong Yin reports administrative support was provided by Zhejiang Province Public Welfare Technology Application Research Project. Xuesong Yin reports a relationship with Wenzhou Public-Welfare Research Project that includes: non-financial support. Xuesong Yin reports a relationship with Key Research and Development Program of Zhejiang Province that includes: non-financial support.Xuesong Yin reports a relationship with Hangzhou Dianzi University that includes: non-financial support. Xuesong Yin reports a relationship with Key Research and Development Program of Zhejiang Province that includes: non-financial support. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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